# Intake Intelligence Agent Swarm: Project Concept

## What it does

This tool analyses an incoming case stream (volume, flow, and workload) and turns it into answers a team can act on: how cases move between stages, how much volume is coming, which categories are surging, whether duplicates exist, what a surge does to backlog and staffing, and what happened this week. Every number comes from a deterministic tool; the model explains results, it never invents arithmetic.

## Questions it answers

1. What is the probability that a given case moves from Stage A to Stage B?
2. How many cases will Stage B receive next month?
3. Which category is growing fastest?
4. If Stage B intake rises 20%, what happens to backlog and staffing?
5. Produce this week's one-page management briefing.

## Agents needed

**Coordinator** (routes, synthesises, critique pass); **Data Agent** (database queries, data-quality checks); **Classification & Duplicate Agent** (categorises case text, finds similar/duplicate cases); **Forecast & Scenario Agent** (volume forecasts, backlog/staffing simulation); **Briefing Agent** (assembles the briefing from a fixed template).

## Design choices

- **Agent Prompting:** role-specific prompts, fixed output schemas, few-shot examples, checklist critic prompt.
- **Agent Orchestration:** Coordinator routes; briefing pipeline fixed (snapshot → growth/anomaly → forecast → briefing → critique); parallel calls; stop-and-ask on missing input; LangChain tool loops.
- **Agent Memory / RAG / MCP:** shared snapshot (short-term); ChromaDB RAG over case texts, past briefings, definitions; MCP canned queries only, with no free-form SQL from the model.
- **Multi-agent Coordination:** divide by capability, hand off via the shared snapshot, report conflicts with scores, degrade gracefully if one agent fails.
- **Safety and Evaluation:** operational analysis only, estimates labelled, public/synthetic data only, every number traced to a tool output, backlog/staffing from formulas (inventory arithmetic, Little's Law); evaluated by forecast MAPE vs. seasonal-naive baseline, spike/duplicate detection rates, and briefing traceability.